Gromov Compactness for Genus 0 Pseudo-Holomorphic Curves

A Formalization Blueprint in Lean 4

Autoformalization Team

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Chapter 1

Status and conventions

The master theorem Theorem 6.7 is a machine-checked *reduction*: it is proved in Lean with no `sorry`, from explicit hypotheses called certificates (Chapter 3). All library declarations depend only on `propext`, `Classical.choice` and `Quot.sound` (census: `tests/PhaseAxiomCensus.lean`). Certificates were audited for satisfiability in `tests/CertificateSanity.lean`; two earlier certificates were shown to be too strong and replaced (Chapter 8). Several certificates are close to, or identical to, the conclusion of the theorem they serve; this is recorded in each node.

Chapter 2

Foundations

Definition 2.1 (Closed tame target)**.** A compact boundaryless smooth manifold M with pointwise nondegenerate skew forms ω_x, closed in charts, an almost-complex structure J with $J^2 = -1$, smooth in charts, and tameness $\omega_x(v, J_x v) > 0$ for $v \neq 0$.

Definition 2.2 (Compatible metric)**.** $g_x(v, w) = \frac{1}{2}(\omega_x(v, J_x w) + \omega_x(w, J_x v))$.

Lemma 2.3 (Metric properties)**.** *g is symmetric, positive definite and J-invariant.*

Proof. Algebra from tameness and $J^2 = -1$. □

Definition 2.4 (J-holomorphic map)**.** u is smooth on s and $du \circ j = J \circ du$ at every point of s.

Lemma 2.5 (Reparametrization invariance)**.** *J-holomorphicity is preserved by biholomorphic reparametrization of the domain.*

Proof. Chain rule for `mfderiv`. □

Definition 2.6 (Möbius frames)**.** Möbius transformations of $\mathbb{CP}^1$, with projective compatibility (`is_moebius`), composition, inverse, and centered stereographic affine rescalings $w \mapsto a + \lambda w$ (`affine_scaling`).

Theorem 2.7 (Three-point transitivity)**.** *Any triple of distinct points of $\mathbb{CP}^1$ can be sent to any other by a unique Möbius map.*

Proof. Adjugate matrices. □

Definition 2.8 (Energy measure and total energy)**.** The push-forward from the stereographic chart of Lebesgue measure with density $\frac{1}{2}|du|_g^2 \cdot 4/(1 + |w|^2)^2$; $E(u)$ is its real total.

Theorem 2.9 (Conformal invariance of energy)**.** *Energy is invariant under Möbius reparametrization.*

Proof. Change of variables with the spherical Jacobian. □

Lemma 2.10 (A nonconstant sphere of finite positive energy)**.** $\mathrm{id} : \mathbb{CP}^1 \to \mathbb{CP}^1$ *is J-holomorphic with* $0 < E(\mathrm{id}) < \infty$.

Proof. Energy density of the identity is identically 1. □

Definition 2.11 (Smooth local convergence)**.** C^∞_{loc} convergence on an open subset of $\mathbb{CP}^1$.

Lemma 2.12 (Closedness of J-holomorphicity)**.** C^∞_{loc} *limits of J-holomorphic maps are J-holomorphic.*

Proof. Pass to the limit in $du \circ j = J \circ du$. □

Lemma 2.13 (Diagonal extraction)**.** *A nested sequence of subsequences has a diagonal subsequence eventually inside each.*

Proof. Cantor diagonal argument. □

Definition 2.14 (Stable sphere tree limit and compactness witness)**.** A finite tree of J-holomorphic spheres with nodal matching and stable ghosts; a witness records a subsequence, frames, C^∞_{loc} convergence, frame collapse and energy convergence.

Theorem 2.15 (Component bounds)**.** $Ph \le B$*; for* $|V| > 1$*,* $G \le P - 2$ *and* $|V| \le 2P - 2$*;* $|V| \le \max(1, 2\lfloor B/h \rfloor - 2)$.

Proof. Handshake lemma and the energy gap packaged in `TargetThresholds`. □

Chapter 3

Certificates (assumed hypotheses)

Proposition 3.1 (Certificate `h_fin`: finite energy)**.** *Each u_n has finite extended energy. (Needed because E is a `toReal`; to be discharged in milestone N1.)*

Proof. Not formalized (planned: Theorem 9.2). □

Definition 3.2 (Universal thresholds)**.** A bundle of constants $\varepsilon_{\mathrm{reg}}, C_{\mathrm{reg}}, \varepsilon_{\mathrm{sph}}, \varepsilon_{\mathrm{neck}}, C_{\mathrm{neck}}, a, h$ together with the estimates they satisfy (ε-regularity, energy gap, neck decay).

Proposition 3.3 (Certificate: existence of thresholds)**.** *For the given target, `TargetThresholds` is inhabited. The master theorem takes a threshold bundle as a parameter. Theorem 4.13 constructs one from Propositions 3.4, 3.5 and 3.7.*

Proof. Not formalized unconditionally. □

Proposition 3.4 (Certificate `h_weitz`: Weitzenböck decomposition)**.** *In stereographic coordinates the energy density e of a J-holomorphic sphere is smooth and $\Delta e = |\cdot|^2 + K$ with $|K| \le be^2$. Essentially equivalent to the conclusion of Lemma 4.2.*

Proof. Not formalized (planned: N3, normal charts). □

Proposition 3.5 (Certificate `h_hit` / `h_hit_exit`: isoperimetric bound and ball exit)**.** $r^2 \le E(u; u^{-1}B_r(x))$ *at boundary-hit radii, and nonconstant spheres exit small balls. Close to the conclusion of Theorem 4.7.*

Proof. Not formalized (planned: N3). □

Proposition 3.6 (Certificate `h_osc`: collar oscillation)**.** *On the collar $A(2r, R/2)$ the oscillation of u is at most $4\sqrt{E(u; A(r, R))}$.*

Proof. Not formalized (planned: N3). □

Proposition 3.7 (Certificate `h_decay`: exponential decay on cylinders)**.** $E(u; A(r_1, r_2)) \le C_0((r/r_1)^a + (r_2/R)^a)E(u; A(r, R))$ *below $\varepsilon_{\mathrm{neck}}$. Identical to the conclusion of Theorem 4.10.*

Proof. Not formalized (planned: N3). □

Proposition 3.8 (Certificate `h_ext`: extension across a puncture)**.** *A finite-energy punctured J-holomorphic sphere has a continuous weak J-holomorphic extension. Identical to the conclusion of Theorem 4.11.*

Proof. Not formalized (planned: N3). □

Proposition 3.9 (Certificate `h_tower`: regularity tower)**.** *Weak J-holomorphic maps are C^n for every n. Implies the conclusion of Theorem 4.5 trivially.*

Proof. Not formalized (planned: N3, Schauder in normal charts). □

Proposition 3.10 (Certificate `h_aa`: Arzelà–Ascoli extraction)**.** *Under local gradient bounds, a subsequence converges in C^∞_{loc}. Identical to the existence part of Theorem 4.6.*

Proof. Not formalized (planned: N3). □

Proposition 3.11 (Certificate `h_conc_extract`: smooth extraction off the singular set)**.** *Along a further subsequence the maps converge in C^∞_{loc} off S to a J-holomorphic u_0, and $\mu = \mu_{u_0} + \sum_{p\in S} \mu\{p\}\delta_p$.*

Proof. Not formalized. □

Definition 3.12 (Genuine defect of a sequence)**.** f is J-holomorphic with bounded finite energy, $\lim_{r\to 0} \lim_n E(f_n; D_r(a)) = m$, and near a the maps eventually stay close to b.

Proposition 3.13 (Certificate `OneLevelCertificate` (Contract B04, sequence-relative))**.** *Every genuine defect of mass $m \geq h$ admits a realized one-level decomposition (subsequence, moving centers and scales, Möbius rescaling frames, frame collapse, C^∞_{loc} convergence off nodes, and genuine child defects).*

Proof. Not formalized (this is the analytic core of bubbling). □

Proposition 3.14 (Certificate `ConcDefectCertificate`)**.** *Every atom of the concentration measure is a genuine defect of a rotated subsequence of $u_{\sigma(n)}$, with its true mass and base value $u_0(p)$.*

Proof. Not formalized. □

Proposition 3.15 (Extension certificates)**.** *Three-point rigidity of homeomorphisms (`h_rigidity`), finiteness of component symmetries (`h_finite_symm`), metrizability of the moduli space (`h_metr`), and the limit extraction certificates for marked curves, moduli space and varying J (`h_extract`).*

Proof. Not formalized. □

Chapter 4

Local analysis

Lemma 4.1 (Scalar mean-value inequality (A01)). *If $e \geq 0$ is C^2 with $\Delta e \geq -be^2$ on D_r and $\int_{D_r} e < \eta$, then $\sup_{D_{r/2}} e \leq Cr^{-2} \int_{D_r} e$.*

Proof. Heinz-type argument in flat coordinates. □

Lemma 4.2 (Bochner inequality (A02)). *$\Delta e \geq -be^2 - ce$ for the energy density of a J-holomorphic sphere.*

Proof. Algebra from Proposition 3.4. □

Theorem 4.3 (ε-regularity (A03)). *Small energy on a disk bounds the energy density on the half disk.*

Proof. Mean-value inequality applied to the Bochner inequality. □

Lemma 4.4 (Flat elliptic estimates (A04)). *Interior Cauchy–Riemann estimates, 2D Morrey–Sobolev embedding, Cauchy–Pompeiu Hölder gain and one Hölder bootstrap step, in flat coordinates.*

Proof. Mathlib's Cauchy integral formula and a near/far splitting. □

Theorem 4.5 (Nonlinear bootstrap (A05)). *Weak J-holomorphic maps are smooth.*

Proof. Immediate from Proposition 3.9. □

Theorem 4.6 (Local smooth compactness (A06)). *Under local gradient bounds a subsequence converges in C^∞_{loc} to a J-holomorphic limit.*

Proof. Extraction from Proposition 3.10; limit is J-holomorphic by Lemma 2.12. □

Theorem 4.7 (Target monotonicity (A08)). *$c r^2 \leq E(u; u^{-1}B_r(x))$ for small r when u leaves $B_r(x)$.*

Proof. From Proposition 3.5. □

Theorem 4.8 (Sphere energy gap (A07)). *Nonconstant J-holomorphic spheres have energy at least $\varepsilon_{\mathrm{sph}} > 0$.*

Proof. Monotonicity at a ball the sphere must exit. □

Theorem 4.9 (Cylinder localization (A09)). *Small annular energy confines the collar image to a small ball.*

Proof. From Proposition 3.6. □

Theorem 4.10 (Exponential decay on cylinders (A10)). *Annular energy decays exponentially away from the collars.*

Proof. Restates Proposition 3.7. □

Theorem 4.11 (Extension across a puncture (A11)). *A finite-energy punctured J-holomorphic sphere extends continuously and weakly J-holomorphically.*

Proof. Restates Proposition 3.8. □

Theorem 4.12 (Removal of singularities (A12)). *The extension is a unique smooth J-holomorphic sphere.*

Proof. Bootstrap of the extension; uniqueness by density of the punctured sphere. □

Theorem 4.13 (Thresholds from analytic certificates (B01)). *Propositions 3.4, 3.5 and 3.7 yield a threshold bundle with $0 < h \leq \min(\varepsilon_{\mathrm{reg}}/2, \varepsilon_{\mathrm{sph}}, \varepsilon_{\mathrm{neck}}/2)$.*

Proof. Take minima of the constants. □

Chapter 5

Concentration and bubbling

Theorem 5.1 (Prokhorov compactness on $\mathbb{CP}^1$)**.** *Bounded sequences of finite measures on $\mathbb{CP}^1$ have weakly convergent subsequences.*

Proof. Diagonal argument over a dense sequence in $C(\mathbb{CP}^1)$ and Mathlib's `RealRMK.rieszMeasure`. □

Theorem 5.2 (Weak compactness of energy measures (C01))**.** *Energy measures of a bounded-energy sequence have a weakly convergent subsequence.*

Proof. From Theorem 5.1. □

Definition 5.3 (Concentration data)**.** A subsequence, a limit measure μ, a finite singular set S and a J-holomorphic limit u_0 off S.

Theorem 5.4 (Existence of concentration data (C02))**.** *Every bounded-energy sequence admits concentration data.*

Proof. Weak compactness, finiteness of S, and Proposition 3.11. □

Theorem 5.5 (Defect identity (C03))**.** $\mu = \mu_{u_0} + \sum_{p \in S} m_p \delta_p$ *with* $m_p \geq \varepsilon_{\mathrm{sph}}$*, and* $E(u_{\sigma(n)}) \to E(u_0) + \sum m_p$*.*

Proof. Finite-measure Portmanteau. □

Lemma 5.6 (Calibrated rescaling (B02))**.** *Hofer's lemma specialized to the plane gives centers and scales with $E(f_n; D_{\lambda_n}(a_n)) = m - h/2$.*

Proof. From `Mathlib.Analysis.Hofer`. □

Theorem 5.7 (Neck vanishing (B03))**.** *Neck energy and neck diameter tend to zero with the stated quantifier order.*

Proof. From cylinder localization and Proposition 3.6. □

Definition 5.8 (One-level decomposition)**.** A bubble v, child points q and masses with $m = E(v) + \sum m_i$, $v(\infty) = b$, and calibration.

Lemma 5.9 (Ghost branching (B05))**.** *A constant bubble has at least two children.*

Proof. Calibration arithmetic. □

Lemma 5.10 (Strict rank decrease (B06))**.** $\lfloor m_i/h \rfloor < \lfloor m/h \rfloor$*; well-founded induction on this rank.*

Proof. Children have mass at most $m - h$. □

Definition 5.11 (Local and realized defect trees)**.** A finite stable tree of J-holomorphic spheres of total energy m attached at b; a realized local defect tree also records the subsequence, per-vertex Möbius frames, transition collapse and C^∞_{loc} convergence off incident nodes.

Theorem 5.12 (Resolution of a genuine defect (B07))**.** *Every genuine defect of mass $m > 0$ is resolved by a realized local defect tree.*

Proof. Energy-rank induction; realized children are grafted sequentially along composed subsequences and telescoped frames (`graft_child_defect_trees_realized`). □

Chapter 6

Assembly and the master theorem

Lemma 6.1 (Global frames (G01))**.** *Rescaling data on a rooted tree determine global Möbius frames with the transition relations.*

Proof. Well-founded recursion on depth. □

Definition 6.2 (Grafted tree and prestable limit data)**.** A tree rooted at u_0 with defect trees attached at S; realized grafted trees and prestable limit data carry the subsequence, Möbius frames, transition collapse and C^∞_{loc} convergence.

Theorem 6.3 (Realized root-sphere grafting theorem)**.** *Grafting realized local defect trees onto the root curve u_0 across all concentration atoms $p \in S$ produces a realized grafted tree with global frame collapse and C^∞_{loc} convergence off nodes.*

Proof. Finite-set induction over $s \subseteq S$, composing subsequences and rotating/telescoping child frames. □

Theorem 6.4 (Prestable limit (G01))**.** *Every bounded-energy sequence of J-holomorphic spheres has prestable limit data.*

Proof. Concentration data, defect identity, realized B07 resolution at each atom of S via Propositions 3.13 and 3.14, and Theorem 6.3. □

Theorem 6.5 (Stabilization (G02))**.** *Unstable ghost roots can be pruned or suppressed, yielding a stable limit.*

Proof. Strong induction on $|V|$. □

Theorem 6.6 (Core compactness (G03))**.** *A subsequence converges to a stable sphere tree limit.*

Proof. Assembly followed by stabilization. □

Theorem 6.7 (Genus-zero Gromov compactness)**.** *Main theorem of the project, as a reduction to the certificates of Chapter 3.*

Proof. Core compactness plus component bounds. □

Chapter 7

Extensions

Theorem 7.1 (Varying almost-complex structures (E01))**.** *Limits of J_n-holomorphic maps with $J_n \to J$ are J-holomorphic (unconditional); compactness for varying J (from Proposition 3.15).*

Proof. Operator-norm squeeze; extraction certificate. □

Theorem 7.2 (Marked compactness (E02))**.** *Compactness for marked stable maps.*

Proof. From the marked extraction certificate. □

Theorem 7.3 (Automorphisms (E04))**.** *Möbius three-point rigidity (unconditional); stability iff finite automorphisms (from Proposition 3.15).*

Proof. Fibrewise finiteness. □

Theorem 7.4 (Moduli space (E05))**.** *$\overline{\mathcal{M}}_{0,k}(M, J, \leq B)$ and its class-A part are compact Hausdorff.*

Proof. Sequential compactness plus metrizability certificate. □

Chapter 8

Certificate audit

Remark 8.1 (Withdrawn certificates and definition repairs). Up to commit `0a3de26` the master theorem assumed `h_one_level` (a decomposition for *every* mass $m \geq h$) and `h_graft_extract` (convergence to *every* grafted tree with arbitrary masses). `tests/CertificateSanity.lean` proves: S2, the former forces a sphere of energy exactly m for each $m \in [h, 2h)$ and so fails on any target missing one such value (e.g. $\mathbb{CP}^1$); S4, the latter is contradictory for sequences with convergent energies and two admissible trees. S1 checks that $u_n = \mathrm{id}_{\mathbb{CP}^1}$ satisfies the non-certificate hypotheses, S3 that Definition 3.12 fails without energy concentration, S5 that `affine_scaling` genuinely translates by its center $a \in \mathbb{C}$, and S6 that `SmoothLocalConvergesOn` holds for the nonconstant identity map on $\mathbb{CP}^1$. Both over-strong certificates were replaced by Propositions 3.13 and 3.14, and `GraftCertificate` was eliminated in favour of the proved theorem Theorem 6.3.

Chapter 9

Normal-chart infrastructure (N1–N3)

Theorem 9.1 (Finite good atlas and `mfderiv` factorization (N1))**.** *A compact manifold has a finite atlas with a positive Lebesgue radius, and* ***`mfderiv`*** *factors through its charts via* ***`tangentCoordChange`****.*

Proof. Finite subcover of precompact coordinate balls and Lebesgue number lemma; chain rule for `extChartAt` and `tangentCoordChange_comp`. □

Theorem 9.2 (Energy density certificate and `h_fin` discharge (N1))**.** *Bounded Riemannian energy density on* $\mathbb{CP}^1$ *implies finite extended energy (Proposition 3.1).*

Proof. Comparison with the finite area of $\mathbb{CP}^1$ under the spherical Jacobian. □

Theorem 9.3 (Quadratic Taylor normal charts, Koszul cancellation and $\mathbb{CP}^1$ model (N2))**.** *Quadratic Taylor normalization* $\Phi_{y_0}(v) = y_0 + v - \frac{1}{2}\Gamma_{y_0}(v, v)$ *yields local diffeomorphisms via the inverse function theorem; composing with atlas charts produces centered charts* Ψ_p *with* $\Psi_p(0) = p$, $d\Psi_p(0) = \mathrm{id}$, *and algebraic Koszul cancellation of the first metric derivatives at* 0. ***`UniformNormalChartFamily`*** *packages the 15 geometric fields (uniform radius, bi-Lipschitz distance comparison, curvature and* C^3 *tensor bounds) and is unconditionally inhabited on* $(\mathbb{CP}^1, \omega_{\mathrm{FS}}, J_{\mathbb{CP}^1})$ *via* ***`concrete_CP1`****.*

Proof. `ContDiffAt.toOpenPartialHomeomorph`, Koszul symmetry cancellation, and explicit stereographic bi-Lipschitz bounds on $\mathbb{CP}^1$. □

Lemma 9.4 (Coordinate Bochner–Weitzenböck and radial monotonicity building blocks)**.** *At the center of a normal chart, vanishing of first metric derivatives implies vanishing of the Koszul form and Christoffel symbols and quadratic energy-density control* $|K_{\mathrm{tensor}}| \le (2C_{g,3} + 4C_{J,3}^2 + 8C_{J,3})e(0)^2$. *In polar coordinates, disk energy splits into circle slices with* $F'(r) = \int_0^{2\pi} e(r,\theta) r\, d\theta$, *and* $F(t) \le \frac{1}{2} t F'(t)$ *integrates to* $(F(r_{\min})/r_{\min}^2) r^2 \le F(r)$.

Proof. Operator-norm bounds on `iteratedFDeriv`, polar integration, and `monotoneOn_of_deriv_nonneg`. □

Lemma 9.5 (Coordinate singularity removal and elliptic bootstrap building blocks)**.** *Iterated Widman hole-filling contraction gives geometric annular decay; finite punctured-sphere energy implies vanishing annular energy tail, vanishing collar oscillation (under* `h_osc`*), and Cauchy filter convergence across the puncture on compact M; the quasilinear Laplacian commutator satisfies $\|N(v, Dv)\| \le (1+C_{J,0})C_{J,1}\|Dv\|^2$; and local Lipschitz gradient bounds yield equicontinuity and separable dense-set Arzelà–Ascoli extraction on subdomains.*

Proof. Dyadic induction, `CompleteSpace.complete` on M, operator-norm bounds, and `tendstoUniformlyOn_o` □

Theorem 9.6 (Phase 1: Unconditional C^0 puncture extension and Möbius-frame Arzelà–Ascoli jet upgrade)**.** *(1) From finite punctured-sphere energy and collar oscillation (*`h_osc`*) alone, any two points in a punctured disk lie in a single collar of energy $< (\varepsilon/5)^2$, yielding vanishing punctured-disk oscillation, unconditional existence of the puncture limit $m_* \in M$, unconditional continuous extension $u_{\mathrm{ext}} = \mathrm{update}\, u\, p\, m_*$, and global* `WeakJHolomorphicOn` *extension across p by punctured-sphere density whenever u_{ext} is C^1 with continuous Cauchy–Riemann defect at p. (2) Under discrete frame topology, any finite list of* `ConvergingFrames` *eventually acts by a single composite $\phi \in$ MoebiusFrame, reducing* `SmoothLocalConvergesOn` *to* `CoreSmoothLocalConvergesOn` *under fixed Möbius frames, while* `tendsto_iteratedFDeriv_writtenInExtCha` *upgrades pointwise chart convergence and local jet bounds to C^∞ smoothness of the limit and convergence of all higher jets* `iteratedFDeriv` $\mathbb{R}$ *k without passing to a further subsequence.*

Proof. Single-collar placement $r = \frac{1}{4}\min(|w_1-a|, |w_2-a|)$, `dense_punctured_sphere`, `tendsto_discrete`, and `tendstoUniformlyOn_iteratedFDeriv_of_pointwise_of_local_jet_bounds`. □

Theorem 9.7 (Local-chart certificate bridge from normal chart certificates (N3))**.** *Given a* `UniformNormalChartFamily` *and conditioned* `UniformNormalChartCertificates` *(where oscillation, decay and puncture removal are conditioned on J-holomorphicity and non-degenerate radii, and proved inhabited on constant curves via* `conditioned_certificates_inhabited_of_all_const`*), the bridge theorems instantiate Propositions 3.4 to 3.10. Unconditional derivation of* `UniformNormalChartCert` *from* `ClosedTameTarget` *for general nonconstant curves remains open.*

Proof. Projections, degenerate-collar splitting, and curvature constant combination. □